

PRACTICAL : WEEK 7

Task - 1:

Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

Code:

```
#include <stdio.h>

#include <stdlib.h>

int global_var = 100;
int global_uninitialized;

void code_function()
{
    printf("Code address      : %p\n", (void *)code_function);
}

int main()
{
    static int static_var = 200;
    int stack_var = 300;
    int *heap_var = malloc(sizeof(int));
    *heap_var = 400;
    printf("Linux Process Memory Addresses\n");
    printf("-----\n");
    code_function();
    printf("Global variable address  : %p\n", (void *)&global_var);
    printf("Global uninitialized    : %p\n", (void *)&global_uninitialized);
    printf("Static variable address  : %p\n", (void *)&static_var);
    printf("Heap variable address    : %p\n", (void *)heap_var);
```

```

printf("Stack variable address   : %p\n", (void *)&stack_var);

printf("\nProcess ID: %d\n", getpid());

free(heap_var);

return 0;
}

```

Output:

```

root@Rahul: ~/memory_lab
root@Rahul:~/memory_lab# nano memory_segments.c
root@Rahul:~/memory_lab# gcc memory_segments.c -o memory_segments
root@Rahul:~/memory_lab# ./memory_segments
Linux Process Memory Addresses
-----
Code address           : 0x5848afbfd1e9
Global variable address : 0x5848afc00010
Global uninitialized    : 0x5848afc0001c
Static variable address : 0x5848afc00014
Heap variable address    : 0x5848cf4772a0
Stack variable address   : 0x7ffd1bb0017c

Process ID: 444
root@Rahul:~/memory_lab#

```

Task - 2:

Make use of the /proc//maps file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

Code:

```

#include <stdio.h>

#include <unistd.h>

int global_var = 100;

int main()
{
    int stack_var = 200;

    printf("Process ID: %d\n", getpid());

    printf("Global variable: %p\n", (void *)&global_var);

    printf("Stack variable: %p\n", (void *)&stack_var);

    printf("\nProcess is running...\n");
}

```

```

printf("Open another terminal and run:\n");

printf("cat /proc/%d/maps\n", getpid());

printf("\nPress Enter to terminate...\n");

getchar();

return 0;
}

```

Output:

```

root@Rahul:~# cat /proc/469/maps
6509d79c5000-6509d79c6000 r--p 00000000 08:30 2538
/root/memory_lab/memory_maps
6509d79c6000-6509d79c7000 r-xp 00001000 08:30 2538
/root/memory_lab/memory_maps
6509d79c7000-6509d79c8000 r--p 00002000 08:30 2538
/root/memory_lab/memory_maps
6509d79c8000-6509d79c9000 r--p 00002000 08:30 2538
/root/memory_lab/memory_maps
6509d79c9000-6509d79ca000 rw-p 00003000 08:30 2538
/root/memory_lab/memory_maps
6509eb16e000-6509eb18f000 rw-p 00000000 00:00 0
[heap]
750b5fa00000-750b5fa28000 r--p 00000000 08:30 42254
/usr/lib/x86_64-linux-gnu/libc.so.6
750b5fa28000-750b5fbb1000 r-xp 00028000 08:30 42254
/usr/lib/x86_64-linux-gnu/libc.so.6
750b5fbb1000-750b5fc00000 r--p 001b1000 08:30 42254
/usr/lib/x86_64-linux-gnu/libc.so.6
750b5fc00000-750b5fc04000 r--p 001ff000 08:30 42254
/usr/lib/x86_64-linux-gnu/libc.so.6
750b5fc04000-750b5fc06000 rw-p 00203000 08:30 42254
/usr/lib/x86_64-linux-gnu/libc.so.6
750b5fc06000-750b5fc13000 rw-p 00000000 00:00 0
750b5fc38000-750b5fc3b000 rw-p 00000000 00:00 0
750b5fc40000-750b5fc42000 rw-p 00000000 00:00 0
750b5fc42000-750b5fc46000 r--p 00000000 00:00 0
[vvar]
750b5fc46000-750b5fc48000 r--p 00000000 00:00 0
[vvar_vclock]
750b5fc48000-750b5fc4a000 r-xp 00000000 00:00 0
[vdso]
750b5fc4a000-750b5fc4b000 r--p 00000000 08:30 42248
/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
750b5fc4b000-750b5fc76000 r-xp 00001000 08:30 42248
/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
750b5fc76000-750b5fc80000 r--p 0002c000 08:30 42248
/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
750b5fc80000-750b5fc82000 t--p 00026000 08:30 42248
/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
750b5fc82000-750b5fc84000 rw-p 00038000 08:30 42248
/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7fff97da2000-7fff97dc3000 rw-p 00000000 00:00 0

root@Rahul:~/memory_lab# nano memory_maps.c
root@Rahul:~/memory_lab# gcc memory_maps.c -o memory_maps
root@Rahul:~/memory_lab# ./memory_maps
Process ID: 469
Global variable: 0x6509d79c9010
Stack variable: 0x7fff97dc10c4

Process is running...
Open another terminal and run:
cat /proc/469/maps

Press Enter to terminate...

```